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Project Report

Smoke and Fire Detection System Project Report

Project Title:

Smoke and Fire Detection System using Arduino UNO

Objective of the Project:

The objective of this project is to design a system that can **detect the presence of fire or smoke** in the surroundings using flame and smoke sensors and then **alert** users through a **buzzer and LEDs**. This type of system is useful in homes, offices, and industries to **prevent fire-related accidents**.

Components Used:

Component	Quantity	Description
Arduino UNO	1	Microcontroller to control the system
Flame Sensor	1	Detects presence of fire
Smoke Sensor (MQ-2)	1	Detects presence of smoke/gas
Buzzer	1	Alerts via sound
Red LED	1	Indicates fire detection
Green LED	1	Indicates smoke detection
Jumper Wires	Many	For connections
Breadboard	1	For temporary connections
USB Cable	1	For programming and power

Working Principle:

- **Flame Sensor** detects fire using infrared light and sends a LOW signal when fire is present.
- **Smoke Sensor (MQ-2)** detects gas/smoke and sends a HIGH signal when smoke is detected.
- The Arduino reads these signals:
 - If **flame is detected**, it turns on the **Red LED** and **buzzer** for 5 seconds.
 - If **smoke is detected**, it turns on the **Green LED** and **buzzer** for 5 seconds.
- Both fire and smoke alerts are **handled independently** but share the same buzzer.
- After 5 seconds, if no further detection is present, the buzzer turns off.

Circuit Diagram

Red LED → Pin 5

Green LED → Pin 3

Buzzer → Pin 4

Flame Sensor → Pin 6

Smoke Sensor → Pin 2

Power: 5V and GND from Arduino are used for sensors and LEDs.

Arduino Code (Main Program):

```
// Pins
#define Flame_sensor 6
#define Smoke_sensor 2
#define Red_led 5
#define Green_led 3
#define Buzzer 4

bool flame_status = false;
bool smoke_status = false;
unsigned long fireDetectedTime = 0;
unsigned long smokeDetectedTime = 0;
bool buzzerActive = false;
bool smokeBuzzerActive = false;

void setup() {
  Serial.begin(9600);
  pinMode(Flame_sensor, INPUT);
  pinMode(Smoke_sensor, INPUT);
  pinMode(Red_led, OUTPUT);
  pinMode(Green_led, OUTPUT);
  pinMode(Buzzer, OUTPUT);

  digitalWrite(Red_led, LOW);
  digitalWrite(Green_led, LOW);
  digitalWrite(Buzzer, LOW);
}

void loop() {
  flame_status = digitalRead(Flame_sensor);
  smoke_status = digitalRead(Smoke_sensor);
  unsigned long currentTime = millis();

  Serial.print("Flame Sensor: ");
  Serial.print(flame_status);
  Serial.print(" | Smoke Sensor: ");
  Serial.println(smoke_status);

  // Fire Detection
  if (flame_status == LOW) {
    digitalWrite(Red_led, HIGH);
    fireDetectedTime = currentTime;
    buzzerActive = true;
```

```

} else {
    digitalWrite(Red_led, LOW);
}

// Smoke Detection
if (smoke_status == HIGH) {
    digitalWrite(Green_led, HIGH);
    smokeDetectedTime = currentTime;
    smokeBuzzerActive = true;
} else {
    digitalWrite(Green_led, LOW);
}

// Buzzer Control
bool fireBuzzerOn = false;
bool smokeBuzzerOn = false;

if (buzzerActive) {
    if (currentTime - fireDetectedTime <= 5000) {
        fireBuzzerOn = true;
    } else {
        buzzerActive = false;
    }
}

if (smokeBuzzerActive) {
    if (currentTime - smokeDetectedTime <= 5000) {
        smokeBuzzerOn = true;
    } else {
        smokeBuzzerActive = false;
    }
}

if (fireBuzzerOn || smokeBuzzerOn) {
    digitalWrite(Buzzer, HIGH);
} else {
    digitalWrite(Buzzer, LOW);
}

delay(100);
}

```

Output:

Condition	Red LED	Green LED	Buzzer
Fire detected only	ON	OFF	ON (5s)
Smoke detected only	OFF	ON	ON (5s)
Both detected	ON	ON	ON (5s)
No fire/smoke	OFF	OFF	OFF

Applications:

- Smart homes and buildings
- Offices and server rooms
- Industrial safety systems
- Kitchen fire alarms
- Gas leakage detection

Future Enhancements:

- Add GSM module to send SMS alerts
- Add IoT/Cloud integration for remote monitoring
- Display readings on LCD or OLED
- Add real-time temperature sensor for better fire detection

Conclusion:

This project successfully demonstrates a low-cost, easy-to-build **fire and smoke detection system** using Arduino. It can serve as a prototype for real-time fire safety and alarm systems, helping prevent disasters in homes and workspaces.